

The transition between fully flooded and starved regimes in EHL

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Abstract

Lubricant starvation of the contact can occur in high-speed oil or grease lubricated bearings resulting in reduced elastohydrodynamic (EHL) film thickness (Lubricat. Sci. 11 (1999) 227). To achieve optimum bearing performance and component life, it is obviously desirable to be able to predict when starvation will occur and the resulting level of surface protection. For both oils and greases, the transition between fully flooded and starved behaviour is determined by lubricant loss and replenishment of the track (ASME Trans. J. Tribol. 120 (1998) 126; Cann PME, Chevalier F, Lubrecht AA. Track depletion and replenishment in a grease lubricated point contact: a quantitative analysis. Proceedings of the 23rd Leeds–Lyon Symposium on Tribology. 1996. p. 405–14). The current paper develops a criterion for the fully flooded and starved transition of an oil-lubricated contact based on four primary factors controlling lubrication level; these are volume of oil, contact dimensions, oil viscosity and speed. Experiments have been carried out using an optical EHL device to investigate the effect of each of these parameters on starvation. The results show how the boundary between the fully flooded and the starved regime and the film thickness in the starved regime depend on these parameters. A single dimensionless parameter (SD), based on replenishment local to the contact, has been established between the operating parameters and the transition from the fully flooded to starved regime.

In this paper, the starvation criterion has been developed for an oil-lubricated contact with a fixed volume of oil present. The next stage is to extend the analysis to grease and the prediction of lubrication failure in bearings.

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1. Introduction

It is essential to know if a bearing is operating in the fully flooded or starved regime, as the dependence of the film thickness on operating conditions is significantly different [1]. The fully flooded regime is characterized by a film thickness, which increases with increasing speed, in accordance with the Hamrock–Dowson rules [4]. In the starved regime, the opposite is true and the film decreases with increasing speed [5]. The lubrication transition was first studied by Wedeven et al. [6] who linked the resulting starved film thickness to the position of the inlet meniscus. Chiu [7] and Kingsbury [8,9] introduced the idea that starvation was controlled by the balance of lubricant supplied to, and

ejected from the track. Chiu [7] also examined the role of the $\eta_0 u$ parameter, in the starved regime. Early experimental work was confined to moderately starved conditions; however, in 1992, Guangteng et al. [5] extended measurements into the heavily starved regime and studied amongst other parameters the influence of the lubricant volume. In more recent studies [2,10,11], a quantitative model predicting the starved film thickness from the amount of oil available in the inlet region was developed.

The consensus of the published work is that starvation is related to the balance of lubricant lost from and supplied to the track. However, most of the work has been concerned with mechanisms of lubricant loss from the track, exceptionally Jacod et al. [12] considered a number of different replenishment mechanisms.

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All of the above work has been confined to liquid lubricant; however, the problem of starvation with grease is far more complex [13]. Grease has a complex time-dependent rheology and thus the amount of lubricant available for track replenishment will vary with time, this problem was recently addressed by Mérieux et al. [14].

This paper consolidates earlier work [2,3] by the authors linking the loss and replenishment mechanisms to provide a model capable of predicting film thickness under any bearing operating condition. In the current study, the model is developed for oil where there is a fixed volume of lubricant available for replenishment. The aim is to develop a parameter that will define safe operating conditions and provide the basis for a more informed choice of lubricant.

2. Model

The model used in this paper is a modification of a more classical replenishment model published previously by the authors [15]. The old model supposed that lubricant replenishment occurred *between* overrollings similar to [7]. However, earlier work by the authors [12] showed that replenishment out of contact was very slow and that rapid flow only occurred in the vicinity of the contact.

The fundamental assumption of both models is that the amount of oil in the track h_{oil} determines the EHL film thickness [2,10,11]. The h_{oil} value depends on the amount of oil present in the vicinity of the track $h_{oil\infty}$. A simplified diagram of the lubricant distribution after overrolling is shown in Fig. 1. Chevalier et al. [2] give the relation between h_c and h_{oil} .

Contrary to Ref. [15], it is assumed that the replenishment is driven by surface tension effects σ_s in the vicinity of the contact. Hence the flow is proportional to σ_s and $h_{oil\infty}$. Similarly to Ref. [15], the replenishment flow is inversely proportional to the base oil viscosity η_0 , the track width a and the velocity u . The ratio of these five parameters is a dimensionless parameter named SD (starvation degree):

$$SD = \frac{\eta_0 u a}{h_{oil\infty} \sigma_s} \quad (1)$$

Typical values of test parameters are given in Table 1.

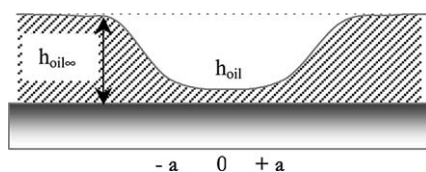


Fig. 1. Schematic diagram of the oil distribution across the track.

Table 1
Typical test values

Oil height ^a	80 μm
Contact width	135 μm
Surface tension	30 mN/m
Critical speed	0.06 m/s
Lubrication viscosity	0.59 Pa s at 30 °C

^a See Section 4.4.

Table 2
Test conditions and base oil properties

Load (N)	10, 20, 30
Temperature (°C)	30, 40, 58
Oil volume (μl)	20, 30, 50
Speed (m/s)	0.01–0.5
Lubrication viscosity (Pa s)	0.59 at 30 °C 0.34 at 40 °C 0.16 at 58 °C

Inserting these values in the SD parameter, one finds that its value is approximately 2 at the transition point. Of the five parameters appearing in SD, four will be investigated in this paper. The effect of each parameter on starvation has been determined in an EHL device.

3. Experimental conditions

The effect of operating parameters on starvation was studied in a ball-on-disc test device supplied by PCS Instruments. Thin film optical interferometry [16] was used to measure central film thickness as a function of speed for a range of test conditions. These are listed in Table 2. The test oil was a high viscosity poly- α -olefin.

The test rig and measurement method have been described in earlier papers [3]. The main difficulty in the current work was developing a method whereby a controlled volume of oil was supplied to the ball track. An automatic pipette that delivers a fixed volume was used to inject lubricant into the contact zone. Great care was taken to ensure that the lubricant was distributed evenly around the rolling track both on the ball and disc before the test was started.

4. Results

4.1. Influence of speed

The first parameter studied is the influence of the velocity u on film thickness, whilst keeping all other parameters fixed. For very small velocities u , the contact will be fully flooded and the film thickness h_c will be proportional to $u^{0.67}$ according to the classical theory of fully flooded lubrication [4]. With increasing speed u , the local replenishment decreases and the con-

tact will no longer be fully flooded. As the lubrication regime becomes fully starved, the film thickness h_c tends to the thickness of the oil layer in the inlet h_{oil} [2]. The thickness of this layer h_{oil} will be proportional to u^{-1} and this leads to the classical starvation curve as a function of speed as shown in Fig. 2.

Under the prevailing conditions, starvation starts around $u = 0.06$ m/s. To study both the fully flooded and starved regime in a more quantitative manner, logarithmic scales are useful as in Fig. 3. Also plotted in this graph are the theoretical curves for the fully flooded and starved regimes, which have slopes of 0.67 and -1.0 , respectively.

From this figure, it can be seen that the experimental film thickness behaves as the theory predicts. Guangteng et al. [5] found that the film thickness stabilized with increasing speed in the starved regime. Fig. 3 does not show this behaviour. However, the film thickness in this region was found to be extremely time dependent. The increasing time to equilibrium can be under-

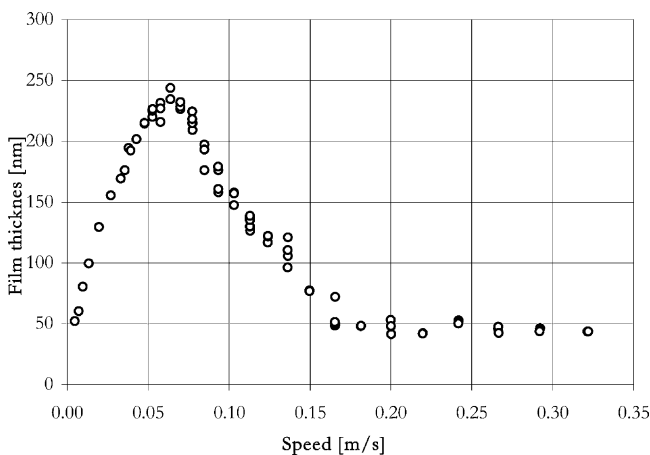


Fig. 2. Film thickness as a function of speed.

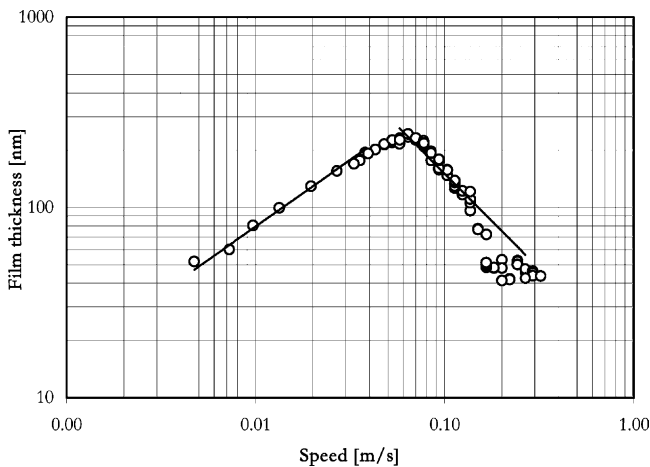


Fig. 3. Film thickness as a function of speed—log–log axes.

stood from a starvation curve without replenishment, as shown in Fig. 8 of [2]. With increasing speed, the time necessary to reach a stable thickness increases exponentially, rendering accurate experiments extremely time consuming.

4.2. Influence of viscosity

A second parameter of interest in both regimes is the viscosity. It is possible to vary the lubricant base oil viscosity by varying the test temperature. For the same volume of oil, the test can thus be carried out for a range of viscosities by progressively increasing the rig temperature.

An example is shown in Fig. 4, where three temperature plots are given, in this case for a 20 μ l lubricant volume. Three distinct curves are obtained with similar maximum film thickness for different speed values.

The differences in the fully flooded regime can be attributed to the viscosity change, as it is the parameter $\eta_0 u$ that determines film formation. Furthermore, the replenishment mechanism also depends on this product (see Eq. (1)). Consequently, when plotting the film thickness as a function of the experimental parameter $\eta_0 u$, one obtains superposition of the three curves (see Fig. 5).

4.3. Influence of load

The load influences the film thickness in two different ways. In the fully flooded regime, the film thickness is proportional to $w^{-0.07}$ so that any increase causes a (very) small film thickness decrease. In the fully starved regime, however, the load influences the replenishment through its effect on the contact radius a and track width. Consequently, the fully flooded film thickness is virtually independent of the load, whereas the onset of starvation is proportional to $w^{-0.333}$, as shown in Fig. 6.

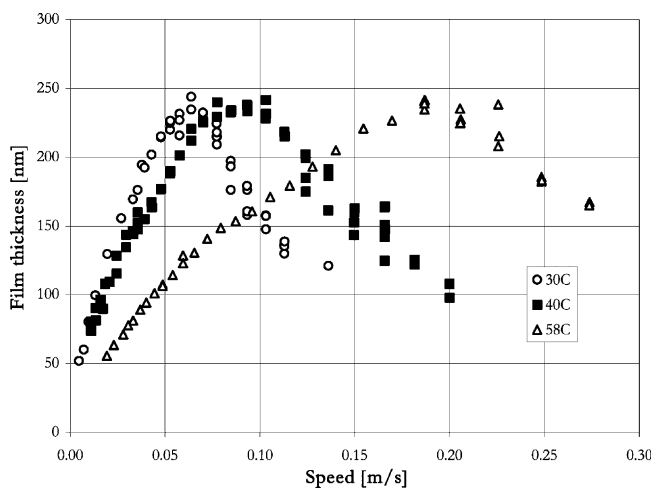


Fig. 4. Film thickness as a function of speed for three temperatures.

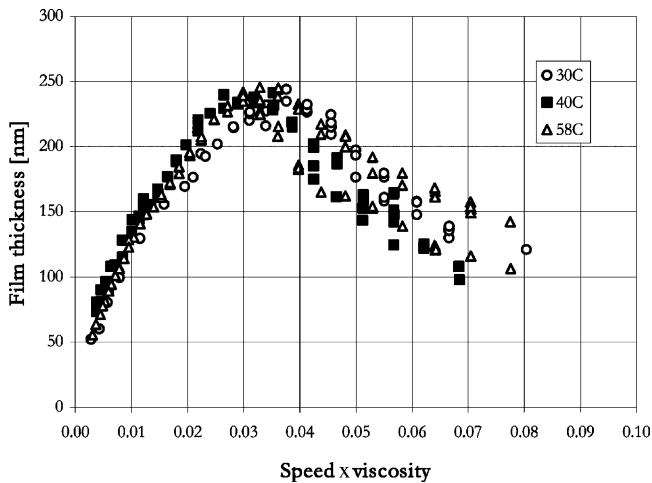


Fig. 5. Film thickness as a function of the product $\eta_0 u$ for three temperatures.

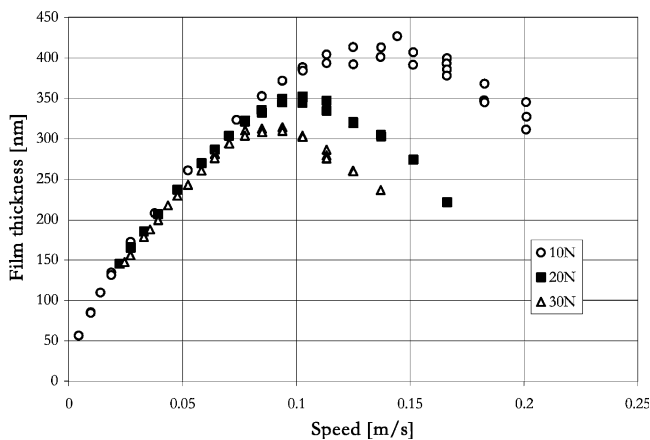


Fig. 6. Film thickness as a function of speed for three values of load.

As the film thickness depends in a different way on w in the two regimes, it is not possible to find a product of u and w that superposes the three curves in both regimes. Therefore, also the vertical axis requires a dimensionless parameter, in this case, the relative film thickness (h_c/h_{eff}), where h_{eff} represents the fully flooded central film thickness. The horizontal axis can be chosen as $uw^{0.333}$, and hence a unique curve is obtained (see Fig. 7).

4.4. Influence of lubricant volume

The volume of lubricant influences the parameter SD through the reservoir height $h_{\text{oil}\infty}$, consequently the central film thickness h_c will also depend on V . As a first approximation, a lubricant layer of constant height and width will give $h_{\text{oil}\infty}$ proportional to V . Assuming a track length of 25 cm and a width of 1 mm, one obtains the values given in Table 1.

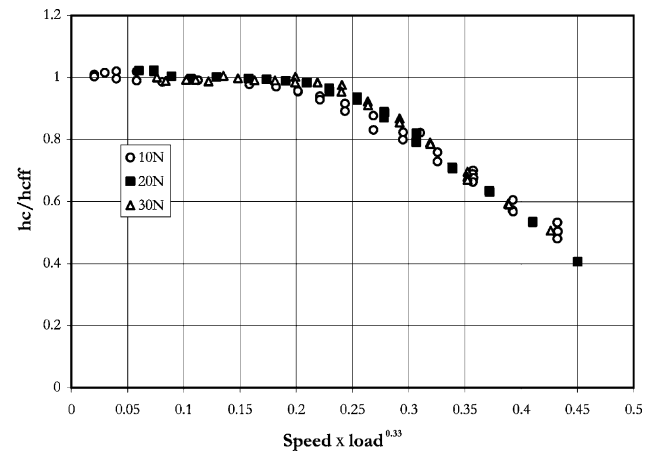


Fig. 7. Dimensionless film thickness as a function of the speed-load parameter.

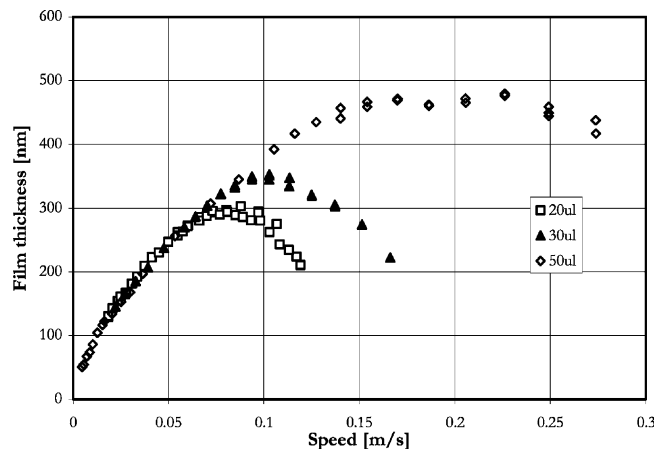


Fig. 8. Film thickness as a function of speed for different volumes of lubricant.

Fig. 8 shows that the volume plays no role in the fully flooded zone, which is the definition of fully flooded lubrication. However, in the starved zone, the film thickness increases with V . Thus, a dimensionless parameter using the ratio $u/h_{\text{oil}\infty}$ should superpose the three curves and this is shown in Fig. 9.

5. Discussion

The data plotted in Figs. 5, 7 and 9 can be represented as a single graph using the SD parameter. The curves follow the fully flooded asymptote ($h_c/h_{\text{eff}} = 1$) up to $SD = 1.5$. Beyond this point, the relative film thickness decreases as $SD^{-1.67}$ as can be shown by classical analysis.

Although the curves have a similar shape, there is some scatter in the results particularly at high SD. This is probably due to errors in both the experimental

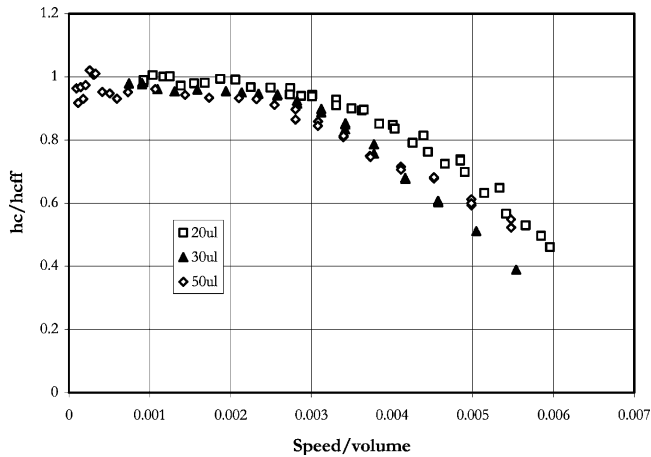


Fig. 9. Dimensionless film thickness as a function of speed/volume.

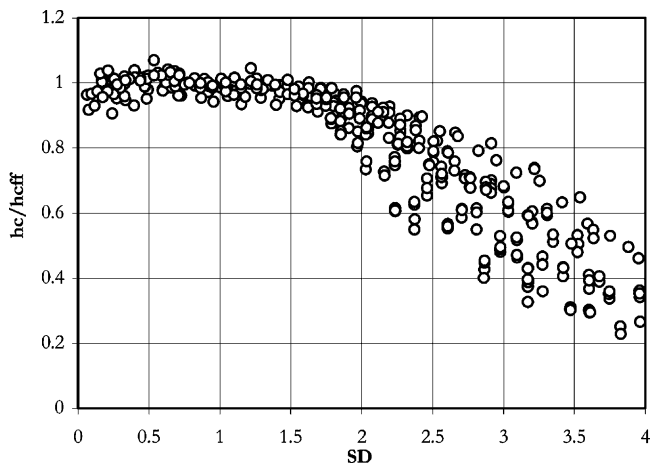


Fig. 10. Relative film thickness as a function of the SD parameter, combining Figs. 5, 7 and 9.

results and the assumptions made in developing the SD parameter. For example, the scatter observed in the viscosity results may stem from poor temperature control. Furthermore, in the model, the surface tension is kept constant throughout the analysis, whereas in reality it decreases typically by 10% between 30 and 50 °C [17].

Finally, it would be expected that the transition between fully flooded and starved behaviour should occur at $SD = 1$. In Fig. 10, this transition is observed between 1.5 and 2. This discrepancy is possibly due to the simplistic geometry assumed for the free oil layer used in the analysis in Section 4.4.

6. Conclusions

The paper establishes the relationship between operational and lubrication parameters and the onset of starvation in EHL contacts. The current analysis devi-

ates from earlier models in one important aspect: lubricant replenishment is now considered as local to the contact rather than occurring on the free surfaces between overrollings. The work has demonstrated the relationships among load, speed, contact size, oil viscosity and volume, and the lubrication regime. A single relationship (SD), based on these five parameters, has been established controlling the onset of starvation. Using this dimensionless parameter, the film thickness can be quantified for different operating conditions. So that for $SD < 1.5$ $h_c/h_{cFF} = 1$, for $SD > 1.5$ $h_c/h_{cFF} = (1.5/SD)^{1.67}$.

The paper has developed a starvation criterion for oil-lubricated contacts. The next challenge is to extend the analysis to the more difficult case of grease lubrication. Grease has a complex time-dependent rheology and the amount of mobile lubricant available for replenishment will change with time and operating conditions. The effect of shear degradation on contact replenishment has been studied in an earlier paper [14]. The next stage of the work therefore is to gather all these elements together to develop a starvation criterion for greases, which will allow prediction of lubrication failure in bearings.

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